

Coordinate Systems in Classical Mechanics

Classical Mechanics – Lecture Notes

1 Cartesian Coordinate System

A coordinate system consists of four basic elements:

- i) choice of origin,
- ii) choice of axes,
- iii) choice of positive direction for each axis,
- iv) choice of unit vectors for each axis.

We illustrate these elements with the Cartesian coordinate system. In three-dimensional space every point P is uniquely labelled by an ordered triple of real numbers (x, y, z) . The ranges of the coordinates are unbounded:

$$-\infty < x < \infty, \quad -\infty < y < \infty, \quad -\infty < z < \infty. \quad (1)$$

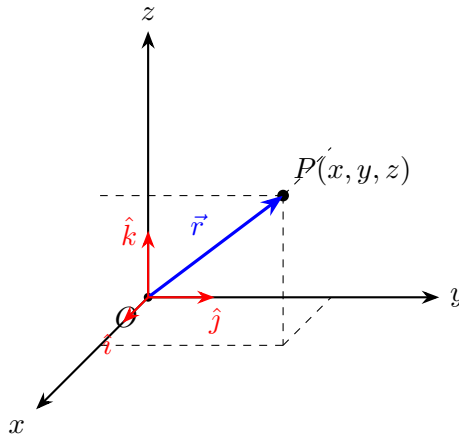


Figure 1: Cartesian coordinate system. The position vector $\vec{r}$ of a point P is resolved into three orthogonal components along the fixed unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$.

At every point we attach a set of three mutually orthogonal unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$. By definition

$$|\hat{i}| = 1, \quad |\hat{j}| = 1, \quad |\hat{k}| = 1.$$

and

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1,$$

$$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0,$$

$$\hat{i} \times \hat{j} = \hat{k}; \hat{j} \times \hat{k} = \hat{i}; \hat{k} \times \hat{i} = \hat{j}.$$

We orient $\hat{i}$ along the direction of increasing x , $\hat{j}$ along the direction of increasing y , and $\hat{k}$ along the direction of increasing z . The position vector of the point $P(x, y, z)$ relative to the origin is then

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}. \quad (2)$$

The scalar coefficients x , y and z are the Cartesian components of $\vec{r}$. The magnitude of position vector is given by

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}.$$

1.1 Choice of positive directions

Conventionally we adopt a right-handed orientation. In two dimensions we draw the positive x -axis to the right and the positive y -axis upward. In three dimensions the positive z -axis points out of the plane when x runs to the right and y runs upward. Nothing prevents us, however, from choosing any other orientation that simplifies a particular problem. A thoughtful choice of axes often turns a difficult calculation into a straightforward one.

2 Infinitesimal Elements in Cartesian Coordinates

2.1 Infinitesimal line element

Consider two neighbouring points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$. The infinitesimal displacement vector that joins them is

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}, \quad (3)$$

where

$$dx = x_2 - x_1, \quad dy = y_2 - y_1, \quad dz = z_2 - z_1.$$

The magnitude of this displacement is the Euclidean distance

$$|d\vec{r}| = \sqrt{dx^2 + dy^2 + dz^2}.$$

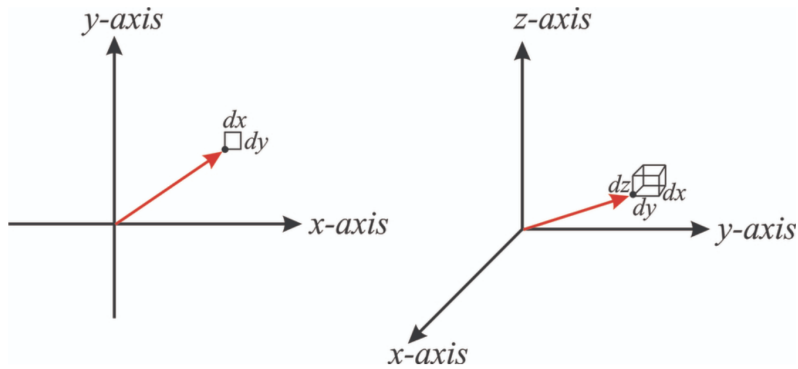


Figure 2: Infinitesimal displacement $d\vec{r}$ between two neighbouring points in the xy -plane.

2.2 Infinitesimal area element

An oriented area element lying in a coordinate plane is the vector

$$d\vec{A}_{\hat{n}} = dA\hat{n}, \quad (4)$$

where $\hat{n}$ is the unit normal to the plane. For example, in the xy -plane, we take $\hat{n} = \hat{k}$, so

$$d\vec{A}_{\hat{k}} = dx\,dy\,\hat{k}.$$

The magnitude $dA = dx\,dy$ is simply the area of the infinitesimal rectangle in that plane.

2.3 Infinitesimal volume element

The volume of an infinitesimal rectangular box with edges parallel to the coordinate axes is

$$dV = dx\,dy\,dz. \quad (5)$$

2.4 Vector $\vec{A}$

Any vector in Cartesian coordinate can be written as

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k},$$

with it's magnitutde

$$|\vec{A}|^2 = A_x^2 + A_y^2 + A_z^2.$$

3 Cylindrical Coordinate System

Whenever a physical problem possesses cylindrical symmetry we gain a decisive advantage by adopting cylindrical coordinates. A generic point P is labelled by the ordered triple (ρ, ϕ, z) , where

- ρ is the perpendicular distance from the z -axis,
- ϕ is the azimuthal angle measured from the positive x -axis in the counterclockwise sense,
- z is the same Cartesian height coordinate.

Thus, the ranges of the cylindrical coordinates are

$$0 \leq \rho < \infty, \quad 0 \leq \phi < 2\pi, \quad -\infty < z < \infty. \quad (6)$$

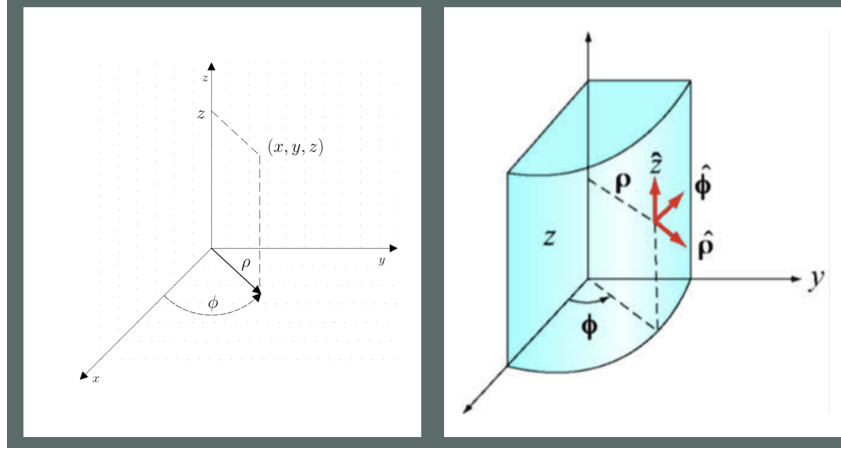


Figure 3: Cylindrical coordinates (ρ, ϕ, z) of a point P . The radial coordinate ρ is the distance from the z -axis, ϕ is the azimuthal angle, and z is the height.

3.1 Unit vectors in cylindrical coordinates

At every point we introduce three orthonormal unit vectors:

- $\hat{\rho}$ points in the direction of increasing ρ (radially outward from the z -axis),
- $\hat{\phi}$ points in the direction of increasing ϕ (tangential to the circle of radius ρ , counterclockwise),
- $\hat{z}$ points in the direction of increasing z and coincides with the Cartesian unit vector $\hat{k}$.

Both $\hat{\rho}$ and $\hat{\phi}$ lie in the plane perpendicular to the z -axis; their orientations change as we move around that plane, while $\hat{z}$ remains constant.

A general vector $\vec{A}$ admits the expansion

$$\vec{A} = A_\rho \hat{\rho} + A_\phi \hat{\phi} + A_z \hat{z}. \quad (7)$$

Its magnitude is

$$|\vec{A}| = \sqrt{A_\rho^2 + A_\phi^2 + A_z^2}. \quad (8)$$

3.2 Orthonormality relations

The cylindrical unit vectors satisfy the same algebraic relations as the Cartesian triad:

$$\hat{\rho} \cdot \hat{\rho} = \hat{\phi} \cdot \hat{\phi} = \hat{z} \cdot \hat{z} = 1, \quad (9)$$

$$\hat{\rho} \cdot \hat{\phi} = \hat{\phi} \cdot \hat{z} = \hat{z} \cdot \hat{\rho} = 0, \quad (10)$$

$$\hat{\rho} \times \hat{\phi} = \hat{z}, \quad \hat{\phi} \times \hat{z} = \hat{\rho}, \quad \hat{z} \times \hat{\rho} = \hat{\phi}. \quad (11)$$

These identities follow at once from the definitions of the three unit vectors and from the right-handed orientation of the basis.

Remark

When the motion is confined to a fixed plane (say $z = \text{const}$) the cylindrical description reduces to the **plane polar coordinate** system (ρ, ϕ) . In that case we retain only the two unit vectors $\hat{\rho}$ and $\hat{\phi}$ that lie in the plane; the third unit vector $\hat{z}$ is normal to the plane.

4 Coordinate Transformations

4.1 From cylindrical to Cartesian coordinates

Given the cylindrical coordinates (ρ, ϕ, z) of a point, the corresponding Cartesian coordinates are obtained by elementary trigonometry:

$$\begin{aligned} x &= \rho \cos \phi, \\ y &= \rho \sin \phi, \\ z &= z. \end{aligned} \quad (12)$$

The inverse transformation reads

$$\begin{aligned} \rho &= \sqrt{x^2 + y^2}, \\ \phi &= \tan^{-1}\left(\frac{y}{x}\right), \\ z &= z. \end{aligned} \quad (13)$$

Because ρ is non-negative we always take the positive square root. The inverse tangent must be evaluated with care: the correct quadrant of ϕ is determined by the signs of both x and y . When $x > 0$ and $y > 0$ one has $0 < \phi < \pi/2$; when both coordinates are negative the angle lies in the third quadrant, and so on.

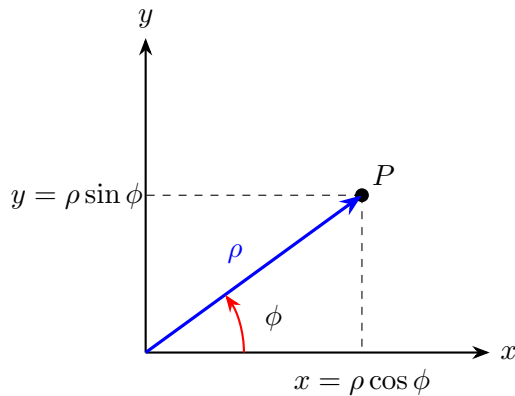


Figure 4: Relation between Cartesian (x, y) and polar (ρ, ϕ) coordinates in the plane.